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MediSyn: A Synergistic multi-agent AI system for IPE in medical domain

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Abstract

The increasing complexity of modern healthcare demands collaborative, inter-professional approaches to ensure comprehensive patient care, yet current AI solutions based on general large language models (LLMs) often fall short in delivering nuanced, profession-specific advice. This project addresses the challenge of integrating Inter-Professional Education/Practice (IPE/IPP) into an AI-driven multi-agent system designed to generate treatment plans for real-life patient cases. We demonstrate that a baseline single LLM solution performs poorly when tasked with generating treatment plans. To overcome these limitations, we propose an innovative approach that simulates real-world healthcare team interactions by incorporating a multi-agent system that dynamically creates agents based on the professions represented in each case study. Rather than relying on a hardcoded set of agents, our system extracts team member roles from real-world inter-professional case studies and assigns each agent the domain-specific knowledge and responsibilities of its corresponding profession. This approach allows, for example, a nurse agent to provide patient care insights and an oncologist agent to deliver targeted cancer treatment advice, closely mirroring real-world inter-professional collaboration. Our experiments show that this specialized, collaborative framework significantly outperforms the general-purpose LLM, paving the way for improved, context-aware treatment planning in clinical settings.

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1 Introduction

The increasing complexity of modern healthcare demands collaborative and interprofessional approaches to ensure comprehensive patient care. The paper "MediSyn: A Synergistic multi-agent AI system for IPE in the medical domain" explores the integration of Inter-Professional Education/Practice (IPE/IPP) principles into an AI-driven framework. It introduces MediSyn, a multi-agent system designed to dynamically create role-specific agents based on real-world case studies, and compares its performance against baseline methods such as single Large Language Models (LLMs) and prompt-tuned LLM pipelines in generating treatment plans. This report provides a comprehensive overview and analysis of the work presented in the paper.

2 Problem Statement

In contemporary healthcare, patient needs are multifaceted, requiring the expertise of various professionals. Interprofessional Education (IPE) and Interprofessional Practice (IPP), as defined by the World Health Organization (WHO), emphasize collaborative learning and teamwork among healthcare workers to enhance communication, reduce medical errors, and improve clinical effectiveness. With the advent of LLMs in medical AI applications, there's growing interest in leveraging their capabilities to support or simulate interprofessional reasoning. However, general-purpose LLMs often lack the role awareness and domain-specific context needed for nuanced clinical decision-making, especially in multi-role environments.

This research investigates how different LLM-based approaches perform in generating treatment plans for real-world IPE case studies. The key question is whether simulating interprofessional dynamics through a role-specialized multi-agent framework, like MediSyn, provides advantages or trade-offs compared to current monolithic approaches.

3 Related Work

The integration of LLMs in healthcare has spurred interest in tasks such as clinical decision support, treatment recommendation, and patient education. While LLMs like GPT-3 and GPT-4 have shown impressive capabilities in understanding and generating medical language, they often lack role-specific contextual awareness for profession-targeted treatment planning. Multi-Agent Systems (MAS) have been explored extensively in healthcare to simulate collaboration by assigning responsibilities to individual agents representing different medical roles. Applications include chronic disease management, prehospital triage, cancer care coordination, and emergency response systems.

Recent advancements have bridged the gap between generative text reasoning and collaborative simulation with LLM-powered MAS frameworks, such as ClinicalAgent for simulating clinical trial coordination and an LLM-based multi-agent system for emergency triage using the Korean Triage and Acuity Scale (KTAS). Furthermore, the integration of hierarchical agents with knowledge graphs and retrieval-augmented generation has enhanced diagnostic reasoning and personalized healthcare services. However, existing work has not yet addressed IPE or IPP case studies using generative AI in a fully dynamic and role-adaptive way. Most prior systems rely on static agent configurations, do not reflect the diversity of real-world care teams, or are not designed for educational or collaborative training contexts. MediSyn builds upon these foundations by dynamically extracting team roles from real IPE case studies and assigning them to profession-specialized agents.

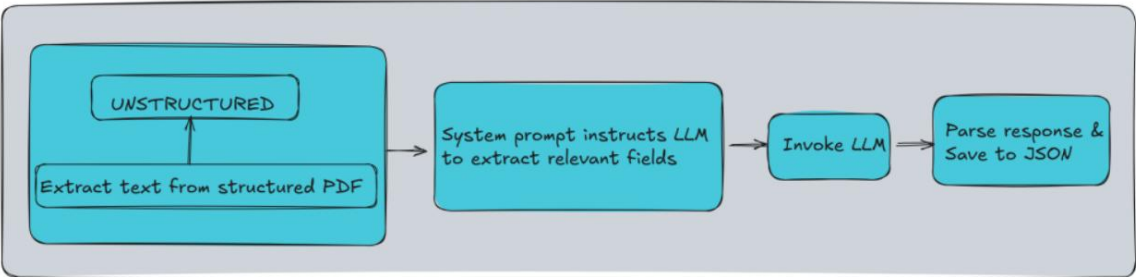
4 Solution and Methodology

The solution section covers all of your contributions (architecture, algorithms, formulas, findings). It explains in detail each contribution, if possible, with figures/schematics. Don't forget that a figure goes a long way towards helping your reader understand your work. For instance, Figure 1 outlines the layers involved in a distributed certification service, and how they articulate together. Nevertheless, a figure must always come with at least one paragraph of explanation. The rule is that anyone should be able to understand your solution from reading the text in this section, even if they skip the figures.

4.1 Data Collection

Data was sourced from 18 publicly available healthcare case studies provided by the American Speech-Language-Hearing Association (ASHA). These case studies serve as authentic real-world examples of interprofessional collaboration in healthcare settings.

4.2 Data Preprocessing



The unstructured case study narratives were processed into a structured JSON format suitable for LLM input. This involved both manual extraction and automated extraction using LLMs, allowing for a comparison of the effectiveness of both approaches. The extracted information was organized into a clean JSON structure including fields such as patient information, summary, team members, background, assessment plan, assessment results, and treatment plan.

Field	Description
patient_info	Includes patient's name, age, and diagnosis.
summary	Summary of the case.
team_members	List of involved professionals.
background	Patient's background.

Field	Description
assessment_plan	Evaluation responsibilities assigned to each professional.
assessment results	Results of diagnostic evaluations.
treatment_plan	The final Interprofessional Practice (IPP) treatment plan co-developed by the team.

4.3 Baseline Approach

Two baseline methods for treatment plan generation were established:

1. **Single LLM pipeline:** The entire case study is passed to a general-purpose LLM (like GPT-4 or Claude) to generate a treatment plan.
2. **Prompt-Tuned LLM pipeline:** The prompt is carefully crafted to simulate specific roles or expectations from the model.

4.4 MediSyn: Proposed Multi-Agent System



MediSyn, a novel synergistic multi-agent architecture, dynamically identifies the healthcare team members mentioned in each case study and creates specialized agents each representing a profession involved in the case. These agents are constrained to their domain knowledge and collaborate sequentially to generate the final treatment plan, thereby closely simulating real-world interprofessional collaboration.

The MediSyn architecture consists of an Agent Builder, Prompt Generator, Agent Instantiation, and Synthesis Agent. The Agent Builder parses the input JSON to extract the list of healthcare roles involved. The Prompt Generator constructs a profession-specific prompt for each role. Individual agents are launched, each constrained to reason within its domain. Finally, the Synthesis Agent integrates the diverse role-specific outputs into a coherent, role-attributed treatment plan.

4.5 Evaluation Metrics

To evaluate and compare the outputs of the Single LLM, Prompt-Tuned LLM, and MediSyn systems, both automated and human-centric evaluation methods were employed:

- BLEU Score
- ROUGE Score

- LLM-as-a-Judge evaluations (for qualitative and contextual judgment)

Each plan is blindly reviewed by an LLM, which is instructed to rate it on a 1–5 scale for each of the following criteria:

- Accuracy: Correctness and clinical relevance of recommendations.
- Plausibility: Whether the recommendations are medically sound and free from hallucinations.
- Specificity: Degree of detail provided (e.g., “hearing screening” vs. “bilateral tympanometry screening with follow-up”).
- Omission: Penalization for missing crucial clinical actions or roles.

4.6 BRIDGE: Platform for IPE Education

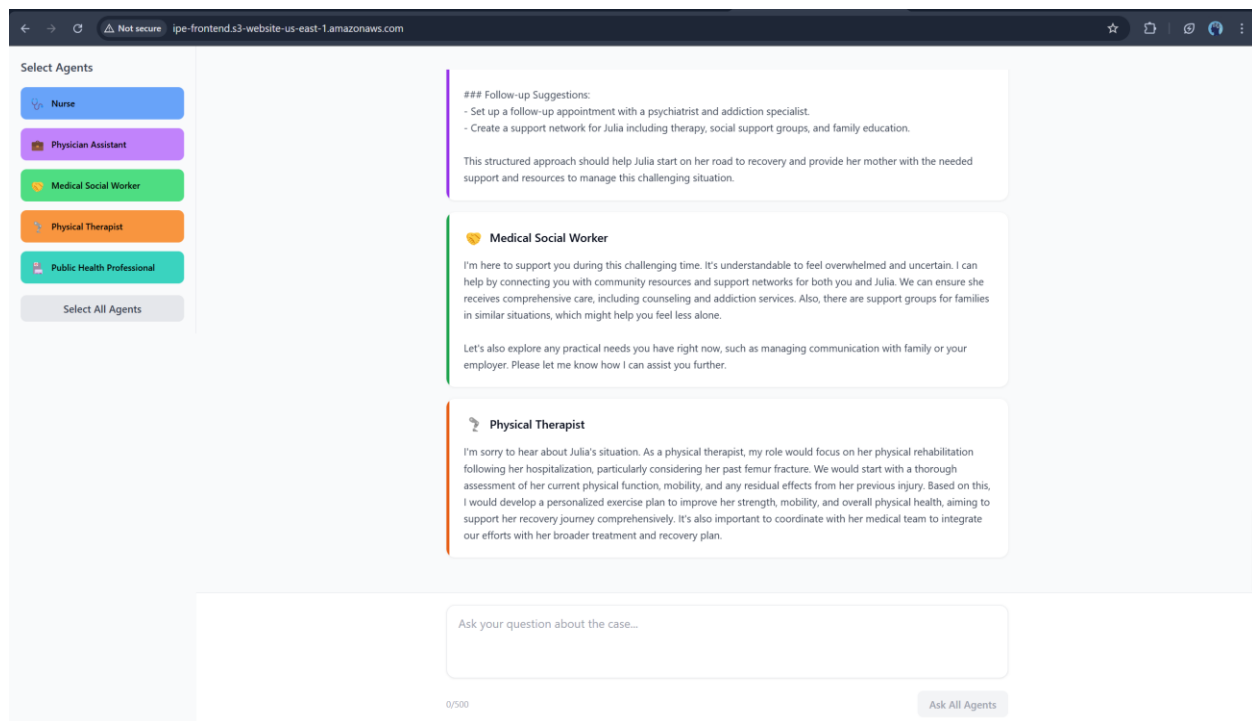


Figure 1: BRIDGE

A separate platform named "BRIDGE" was developed as a learning tool for IPE education. Bridge utilizes five pre-defined agents representing:

- Nurse
- Physician Assistant
- Medical Social Worker
- Physical Therapist
- Public Health Professional

This platform enables students to learn from interactions with these selected agents, simulating a collaborative healthcare environment.

5 Results and Discussion

The experiments aimed to evaluate the effectiveness of different language model configurations in generating interprofessional treatment plans. Three approaches were compared: single general-purpose LLM, prompt-tuned LLM, and MediSyn. The objective was to assess the degree to which each method could produce outputs that were accurate, plausible, specific to team member roles, and complete in terms of clinical content coverage.

All three methods were tested across three LLMs (GPT-40, Claude 3.5 Sonnet, and LLAMA 3.2) under two data settings: manually preprocessed into structured JSON, and LLM-extracted structured data. Results were reported across both automatic evaluation metrics (BLEU, ROUGE) and human-centered criteria (accuracy, plausibility, specificity, and omission rate).

5.1 Prompt Engineering

Prompt engineering was integral to four key stages of the pipeline: 1. Structured Data Extraction, 2. Treatment Plan Generation, 3. Multi-Agent Role Synthesis, and 4. LLM-as-a-Judge Evaluation. These prompt designs enabled more controlled and context-sensitive outputs from the LLMs, facilitating both accurate data ingestion, high-quality generation, and robust evaluation.

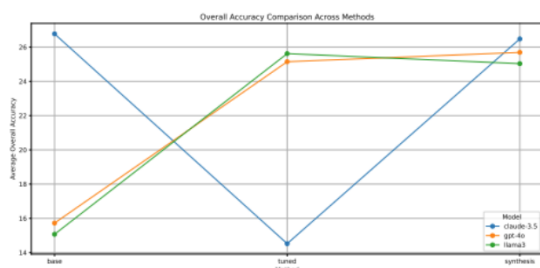
5.2 Data Preparation Comparison

The study compared outputs generated by GPT-40, Claude 3.5 Sonnet, and LLaMA 3.2 in extracting structured information from unstructured medical case study PDFs. GPT-40 most closely matched the manual version in both structure and content. Claude and LLaMA tended to be more verbose or inferential in their outputs. Across the fields examined (patient_info, summary, team_members, background, and treatment_plan), all three LLMs demonstrated the ability to extract structured information with varying levels of fidelity.

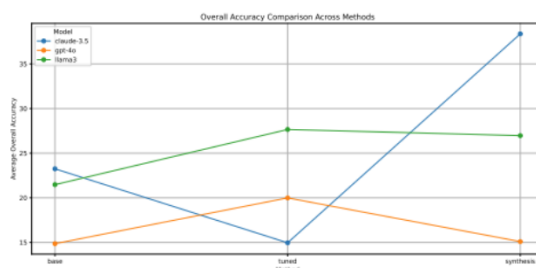
5.3 Treatment Plans Comparison

The final treatment plans generated by each language model under the three configurations were qualitatively compared. The LLama MediSyn output presented the most professional and polished plan. The Claude MediSyn outputs closely mirrored the prompt-tuned version but with slight improvements in transitions and alignment across roles. The MediSyn output from GPT-40 was well-structured, detailed, and clearly aligned with IPE principles. Across all three models, there was a consistent progression in treatment plan quality from base prompt to prompt-tuned and MediSyn.

5.4 LLM vs. Prompt-Tuned LLM vs. MediSyn



(a) Manual JSON



(b) LLM JSON

The study compared three generation paradigms: Base Agent (single LLM), Prompt-Tuned, and MediSyn under both manually extracted and LLM-extracted JSON conditions. Claude 3.5 showed the strongest

performance under the MediSyn configuration on human-aligned metrics such as accuracy, specificity, and plausibility. For GPT-40, prompt tuning led across most metrics. For LLAMA 3.2, prompt-tuned output achieved the highest accuracy and plausibility, while MediSyn improved specificity

6 Discussion

The discussion section of the "MediSyn: A Synergistic Multi-agent AI System for IPE in the Medical Domain" article focuses on several key points:

- **Effectiveness of MediSyn:** The article highlights that MediSyn consistently improves the structural integrity, role attribution, and depth of treatment plans compared to single-LLM and prompt-tuned methods. This is particularly evident for models like Claude 3.5 and LLaMA 3.2. GPT-40 also shows enhanced interprofessional coherence within the MediSyn framework.
- **Qualitative vs. Quantitative Evaluation:** A key discussion point is the contrast between qualitative and quantitative evaluation outcomes. While qualitative assessment favors MediSyn for its clarity and IPE alignment, quantitative metrics like BLEU and ROUGE don't always reflect these strengths. This raises questions about the reliability of traditional lexical metrics for evaluating clinical reasoning and suggests a need for more human-aligned criteria.
- **Limitations of MediSyn:** Several limitations are acknowledged:
 - Using GPT-40 as an LLM-based evaluator is a proxy for human judgment and may not fully capture clinical appropriateness.
 - The sample size of 15 case studies is modest for large-scale benchmarking.
 - The analysis focused on three LLMs without fine-tuning, and domain-adapted versions might yield different results.
 - The modularity of MediSyn introduces challenges in crafting role prompts and ensuring synthesis quality, with potential propagation of errors from auto-extracted input.

Potential of MediSyn: Despite these limitations, MediSyn is seen as a promising foundation for simulating collaborative clinical reasoning, offering value for AI-driven medical education and future interprofessional support systems.

7 Conclusion

We introduced MediSyn, a multi-agent generation framework that simulates interprofessional reasoning for medical treatment planning. By dynamically instantiating role-specific agents and synthesizing their outputs, MediSyn produces structured, collaborative plans aligned with IPE/IPP practices. Evaluated on 15 diverse case studies, MediSyn demonstrated strong performance in producing clinically coherent and role-attributed outputs, particularly when paired with models like Claude 3.5 and LLaMA 3.2. While prompt tuning improved alignment in many cases, MediSyn consistently enhanced structure and interprofessional clarity—especially under noisy, LLM-prepared inputs.

Our findings underscore the limitations of surface-level metrics and highlight the value of human-aligned evaluation criteria such as specificity and plausibility. As LLMs continue to evolve, frameworks like MediSyn offer a promising foundation for AI-assisted education and team-based clinical reasoning, with opportunities for further refinement through expert evaluation and domain-specific tuning.

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